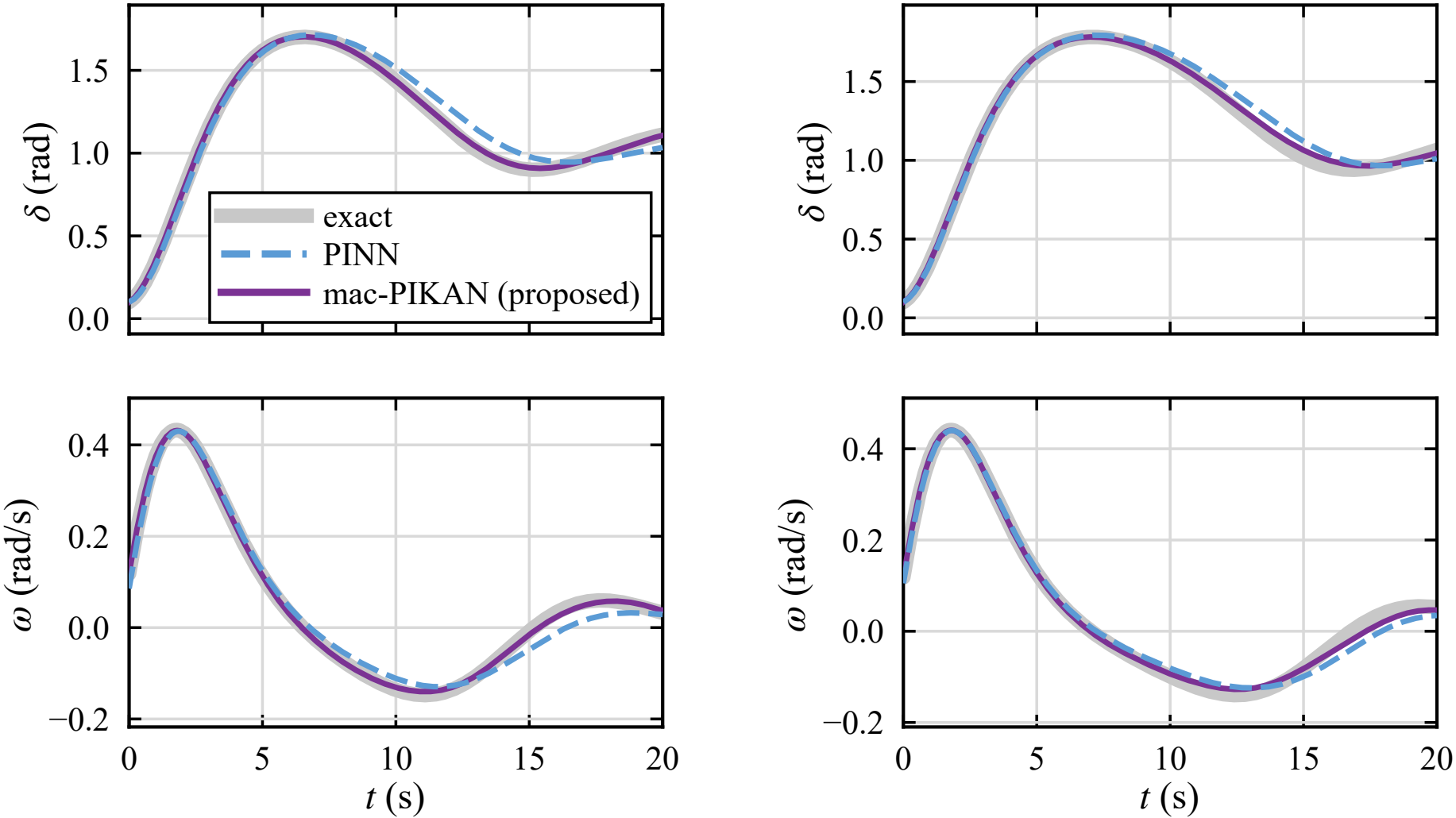
δ (rad)
1.5
1.0
0.5
0.0
exact
PINN
mac-PIKAN (proposed)
ω (rad/s)
0.4
0.2
0.0
−0.2
0
5
10
15
20
t (s)